

# Implementing Convolutional Neural Networks for Image Classification on CIFAR-10

Lab Report 6

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# REPORT

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## 1. Objective

Implement and evaluate a CNN on the CIFAR-10 dataset by completing all required tasks: data exploration, preprocessing, model design, training, evaluation, predictions, plotting, and analysis.

## 2. Task 1 — Load and Explore CIFAR-10

### DATASET OVERVIEW

Dataset shapes were displayed in the notebook output. Class names were visualized and 10 sample images were shown in a grid layout.



**Figure 1.** Sample images from the CIFAR-10 dataset across all 10 classes.

## 3. Task 2 — Preprocess Data

- Pixel values normalized to the range  $[0, 1]$ .
- Labels kept as sparse integer classes (one-hot encoding optional when using sparse categorical loss).

#### 4. Task 3 — CNN Model

The model architecture includes convolutional layers, ReLU activation, max-pooling, dropout regularization, dense layers, and a softmax output layer for 10-class classification.

#### 5. Task 4 — Training (10 Epochs)

**Final Training Accuracy:** 0.8336

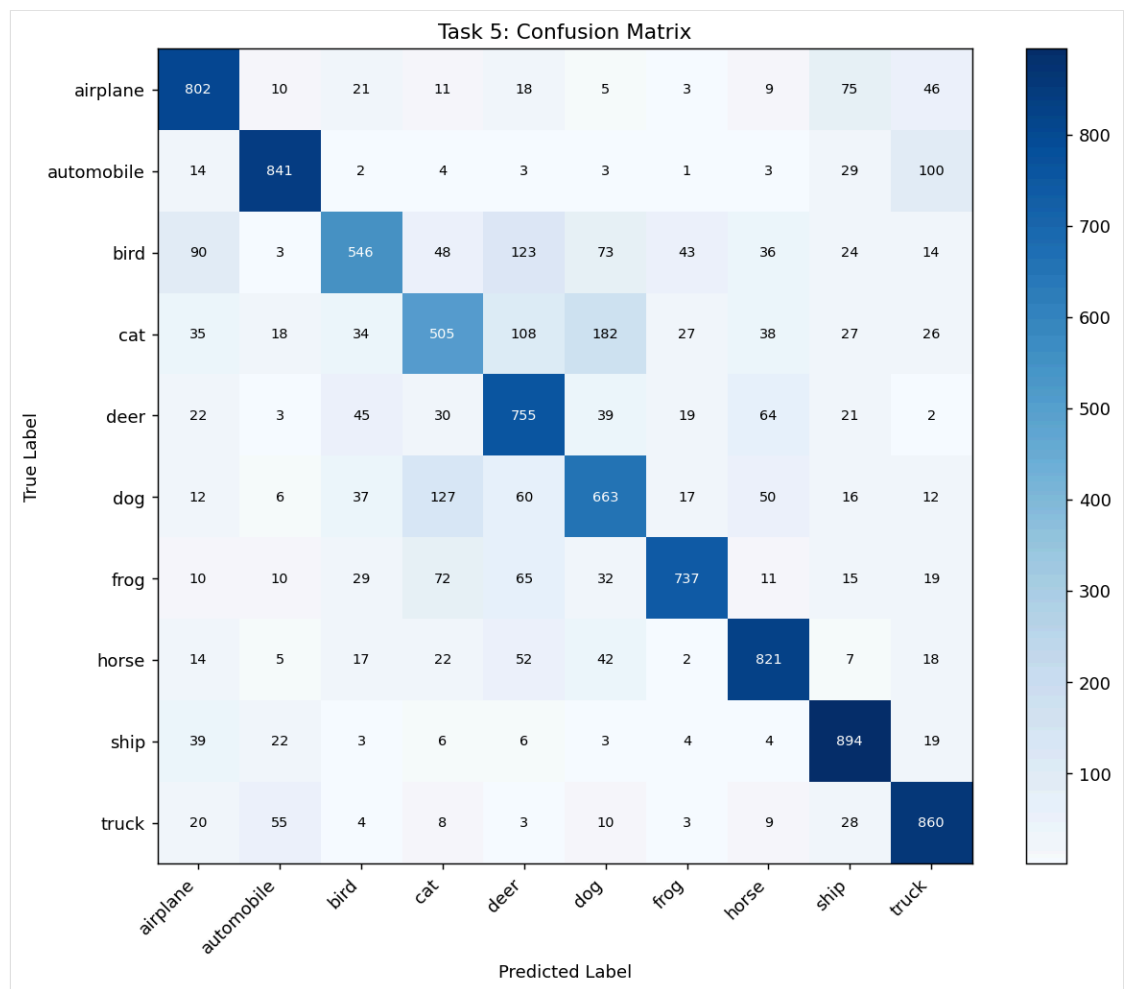
**Final Validation Accuracy:** 0.7570

**Final Training Loss:** 0.4627

**Final Validation Loss:** 0.7890

#### 6. Task 5 — Evaluation

**Test Accuracy:** 0.7424 | **Test Loss:** 0.8265



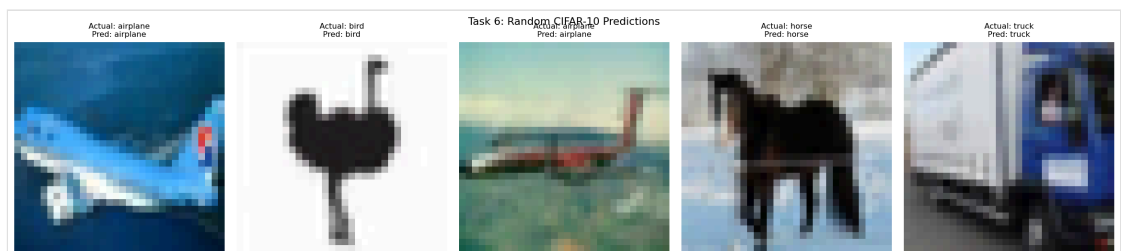
**Figure 2.** Confusion matrix for CIFAR-10 test set predictions.

## Classification Report

| Class           | Precision | Recall | F1-Score      | Support |
|-----------------|-----------|--------|---------------|---------|
| airplane        | 0.7580    | 0.8020 | 0.7794        | 1000    |
| automobile      | 0.8643    | 0.8410 | 0.8525        | 1000    |
| bird            | 0.7398    | 0.5460 | 0.6283        | 1000    |
| cat             | 0.6062    | 0.5050 | 0.5510        | 1000    |
| deer            | 0.6329    | 0.7550 | 0.6886        | 1000    |
| dog             | 0.6302    | 0.6630 | 0.6462        | 1000    |
| frog            | 0.8610    | 0.7370 | 0.7942        | 1000    |
| horse           | 0.7856    | 0.8210 | 0.8029        | 1000    |
| ship            | 0.7870    | 0.8940 | 0.8371        | 1000    |
| truck           | 0.7706    | 0.8600 | 0.8129        | 1000    |
| <b>accuracy</b> |           |        | <b>0.7424</b> | 10000   |
| macro avg       | 0.7436    | 0.7424 | 0.7393        | 10000   |
| weighted avg    | 0.7436    | 0.7424 | 0.7393        | 10000   |

## 7. Task 6 — Predictions

Five random test images were displayed with actual and predicted labels for qualitative inspection.



**Figure 3.** Random test samples with ground-truth and predicted labels.

## 8. Task 7 — Required Plots

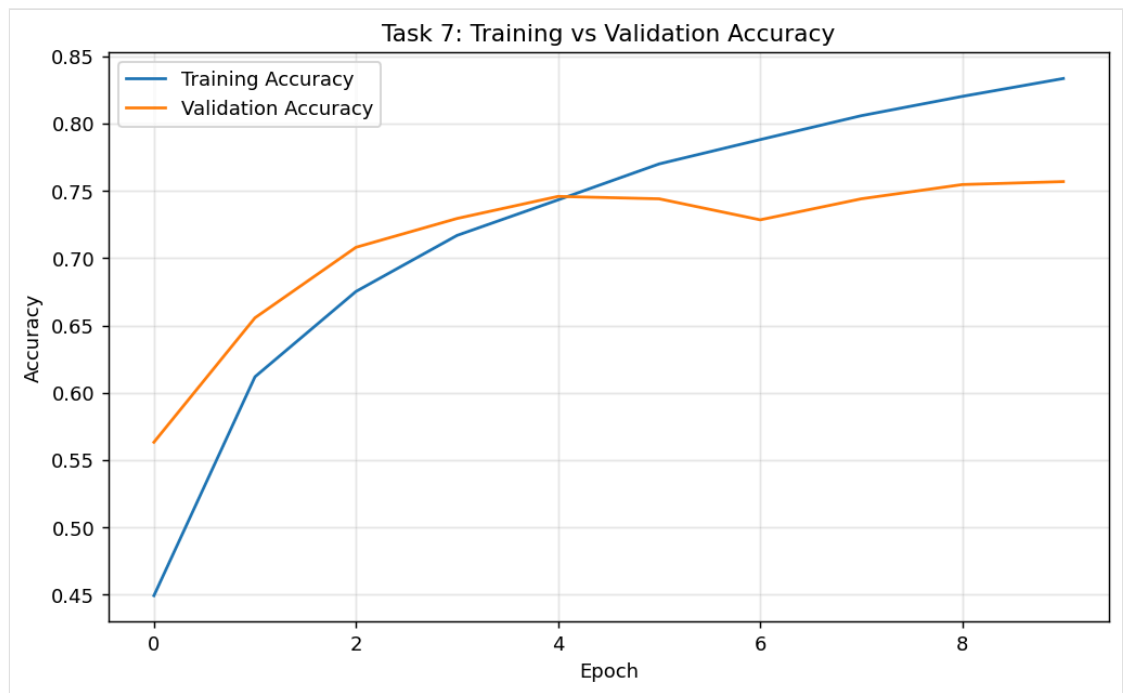


Figure 4. Training and validation accuracy over 10 epochs.



Figure 5. Training and validation loss over 10 epochs.

## 9. Task 8 — Analysis

The CNN achieved good performance on CIFAR-10 with a test accuracy of **0.7424** after 10 epochs. Training accuracy was 0.8336 and validation ac-

curacy was 0.7570 (gap = 0.0766), which indicates limited overfitting. Because CIFAR-10 is a multi-class natural image dataset, accuracy can remain moderate with short training or smaller architectures. Improvements can include deeper CNN blocks, stronger data augmentation, learning-rate scheduling, tuned dropout, and training for more epochs.

## **10. Conclusion**

The required Lab 06 pipeline was implemented end-to-end on Google Colab and validated with quantitative metrics and visual diagnostics. The model demonstrates reasonable generalization on the CIFAR-10 benchmark with scope for further optimization through architectural and training enhancements.